Chapter 4: Space-Filling Designs

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2025-12-31

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1 Introduction

This chapter covers space-filling designs including clustering-based designs, maximin designs, Latin hypercube designs, and MaxPro designs.

Citation: V. R. Joseph (2026). *Experimental Design for Data Science and Engineering.* CRC Press.

```
library(SFDesign)
library(deldir)
library(MASS)
library(plotrix)
library(MaxPro)
```

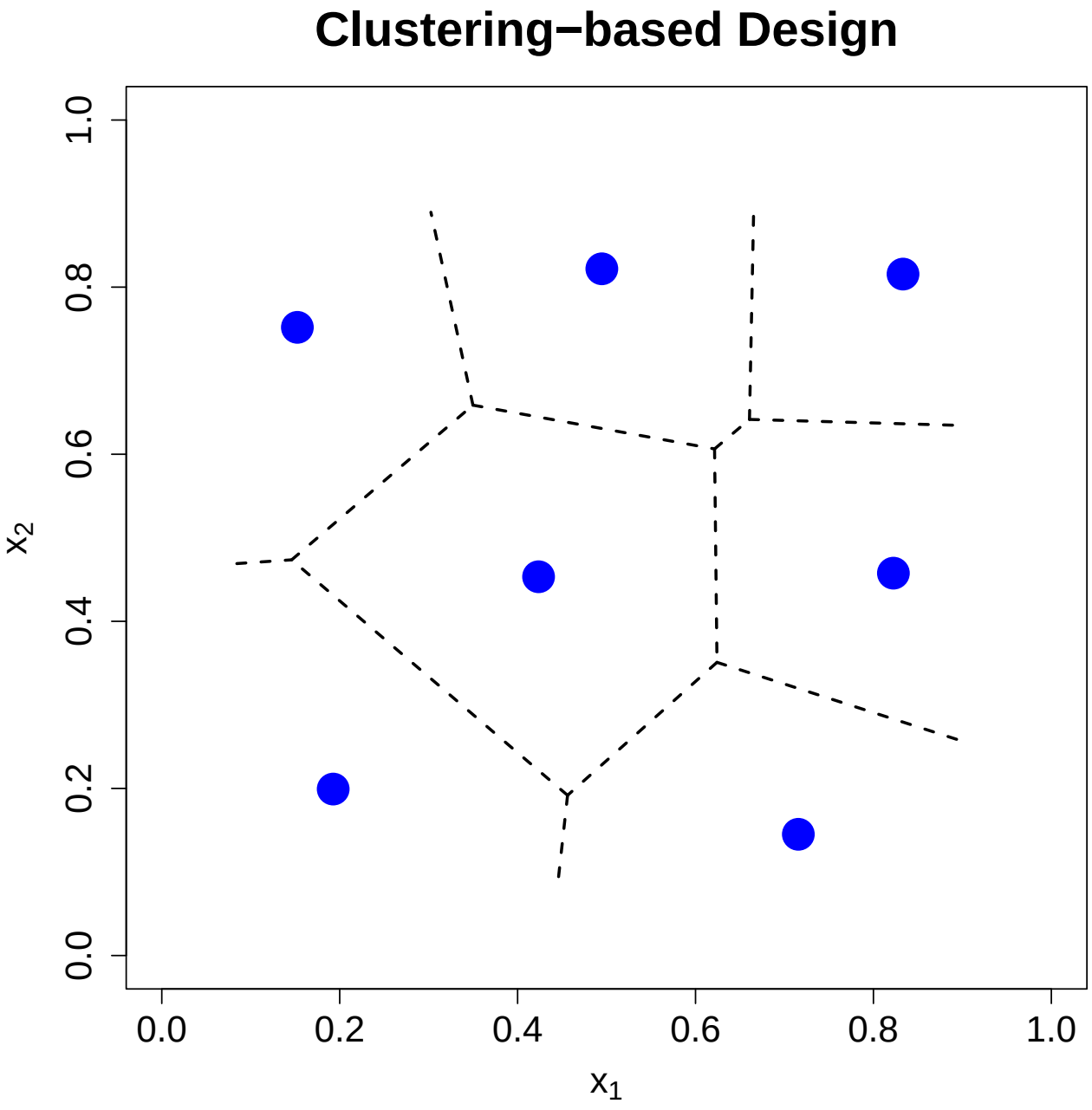
2 Clustering-based Designs

```
p <- 2
n <- 7

D <- clustering.design(n, p)$design

par(mar = c(5, 5, 4, 2))
```

```r
plot(D, xlim = c(0, 1), ylim = c(0, 1), main = "Clustering-based Design",
     xlab = expression(x[1]), ylab = expression(x[2]),
     cex.main = 2, cex.lab = 1.5, cex.axis = 1.5, pch = 16, col = "blue", cex = 3)
vor <- deldir(D[, 1], D[, 2])
plot.deldir(vor, add = TRUE, wlines = "tess", xlim = c(0, 1), ylim = c(0, 1),
            showpoints = FALSE, lwd = 2)
```



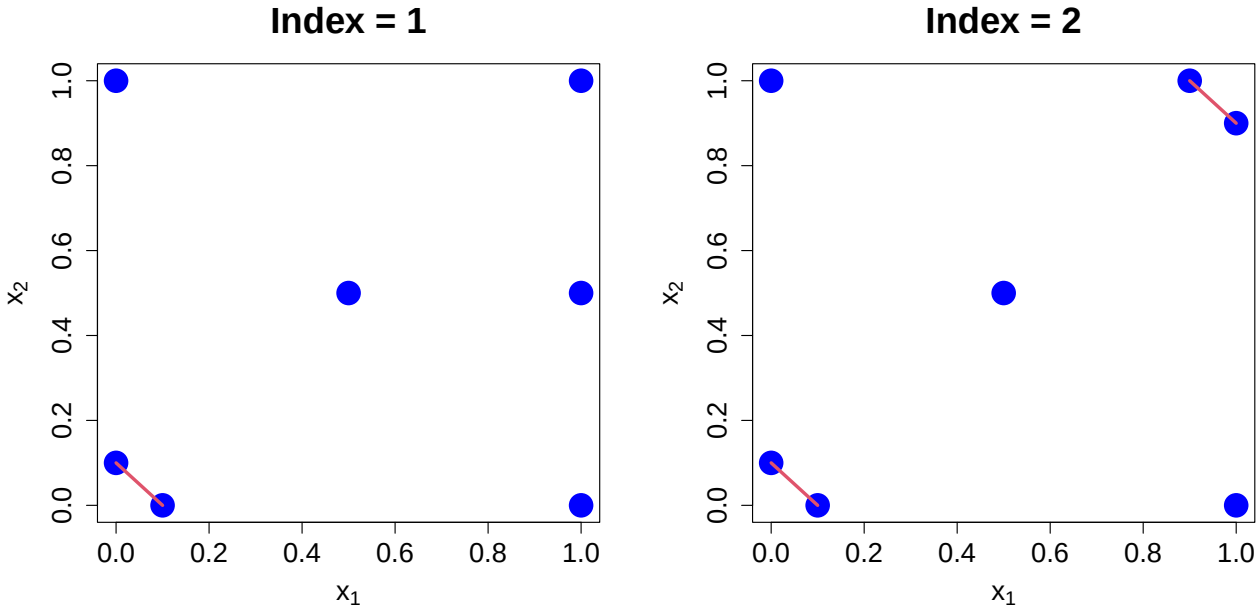
3 Maximin Designs

```r
p <- 2
n <- 7
```

```r
par(mar = c(5, 5, 4, 2))
par(mfrow = c(1, 2))

# Example 1: Index 1
D1 <- matrix(c(0, 0.1, 0.1, 0, 0, 1, 1, 1, 1, 0.5, 1, 0, 0.5, 0.5),
             nrow = n, ncol = p, byrow = TRUE)
plot(D1, xlim = c(0, 1), ylim = c(0, 1), xlab = expression(x[1]),
     ylab = expression(x[2]), cex.lab = 1.5, cex.axis = 1.5, pch = 16,
     col = "blue", cex = 3, main = "Index = 1", cex.main = 2)
segments(0, 0.1, 0.1, 0, col = 2, lwd = 3)

# Example 2: Index 2
D2 <- matrix(c(0, 0.1, 0.1, 0, 0, 1, 0.9, 1, 1, 0.9, 1, 0, 0.5, 0.5),
             nrow = n, ncol = p, byrow = TRUE)
plot(D2, xlim = c(0, 1), ylim = c(0, 1), xlab = expression(x[1]),
     ylab = expression(x[2]), cex.lab = 1.5, cex.axis = 1.5, pch = 16,
     col = "blue", cex = 3, main = "Index = 2", cex.main = 2)
segments(0, 0.1, 0.1, 0, col = 2, lwd = 3)
segments(0.9, 1, 1, 0.9, col = 2, lwd = 3)
```

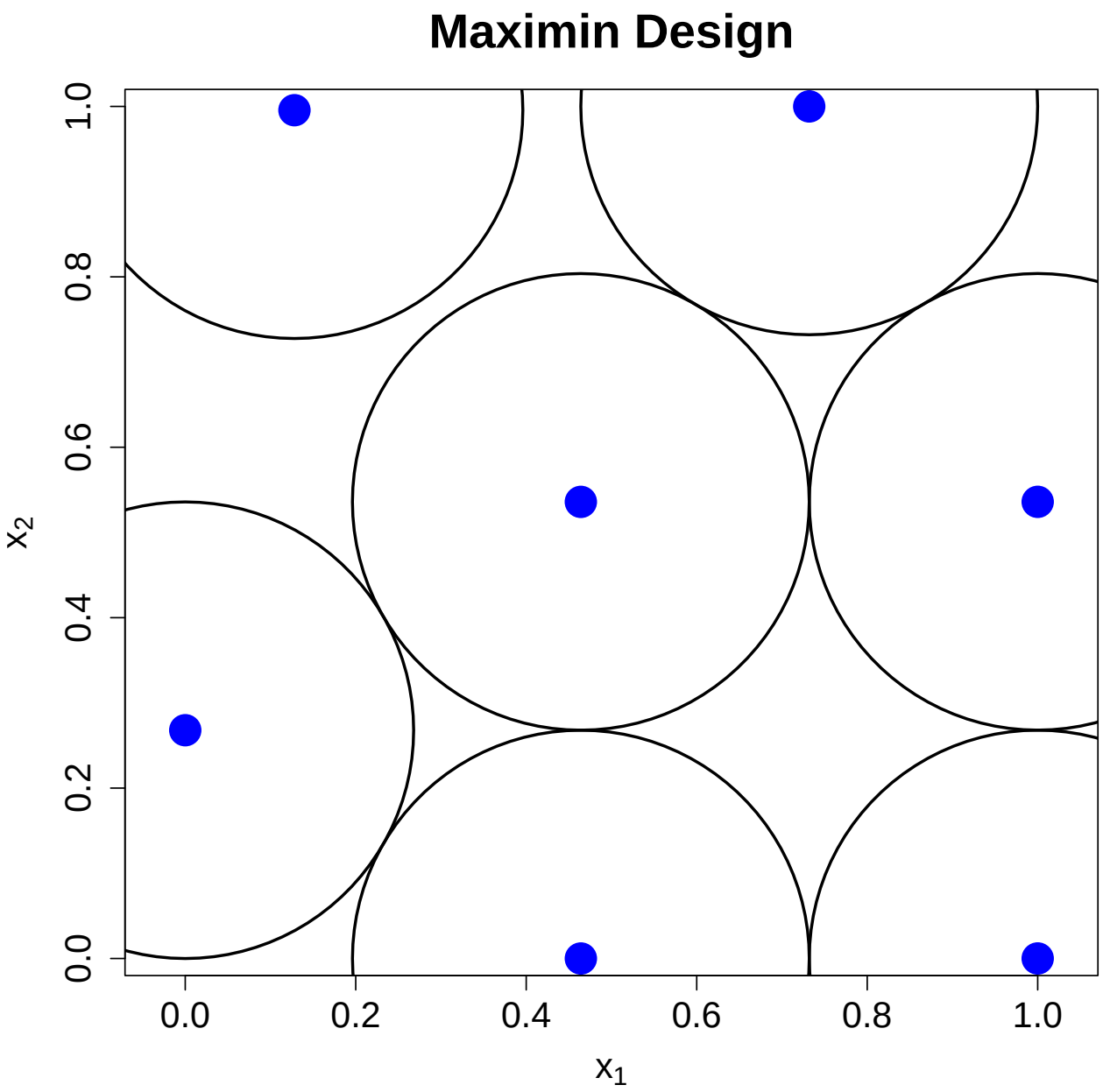



```r
par(mfrow = c(1, 1))
```

3.1 Optimized Maximin Design

```r
set.seed(1)
D <- maximinLHD(n, p)$design
D <- maximin.optim(D, sa = TRUE, find.best.ini = TRUE)$design
r <- min(dist(D)) / 2

eqscplot(D[, 1], D[, 2], xlab = expression(x[1]), ylab = expression(x[2]),
         xlim = c(0, 1), ylim = c(0, 1), pch = 16, col = "blue", cex = 3,
         cex.main = 2, cex.lab = 1.5, cex.axis = 1.5, main = "Maximin Design")
for (i in 1:n) draw.circle(D[i, 1], D[i, 2], radius = r, lwd = 2)
```



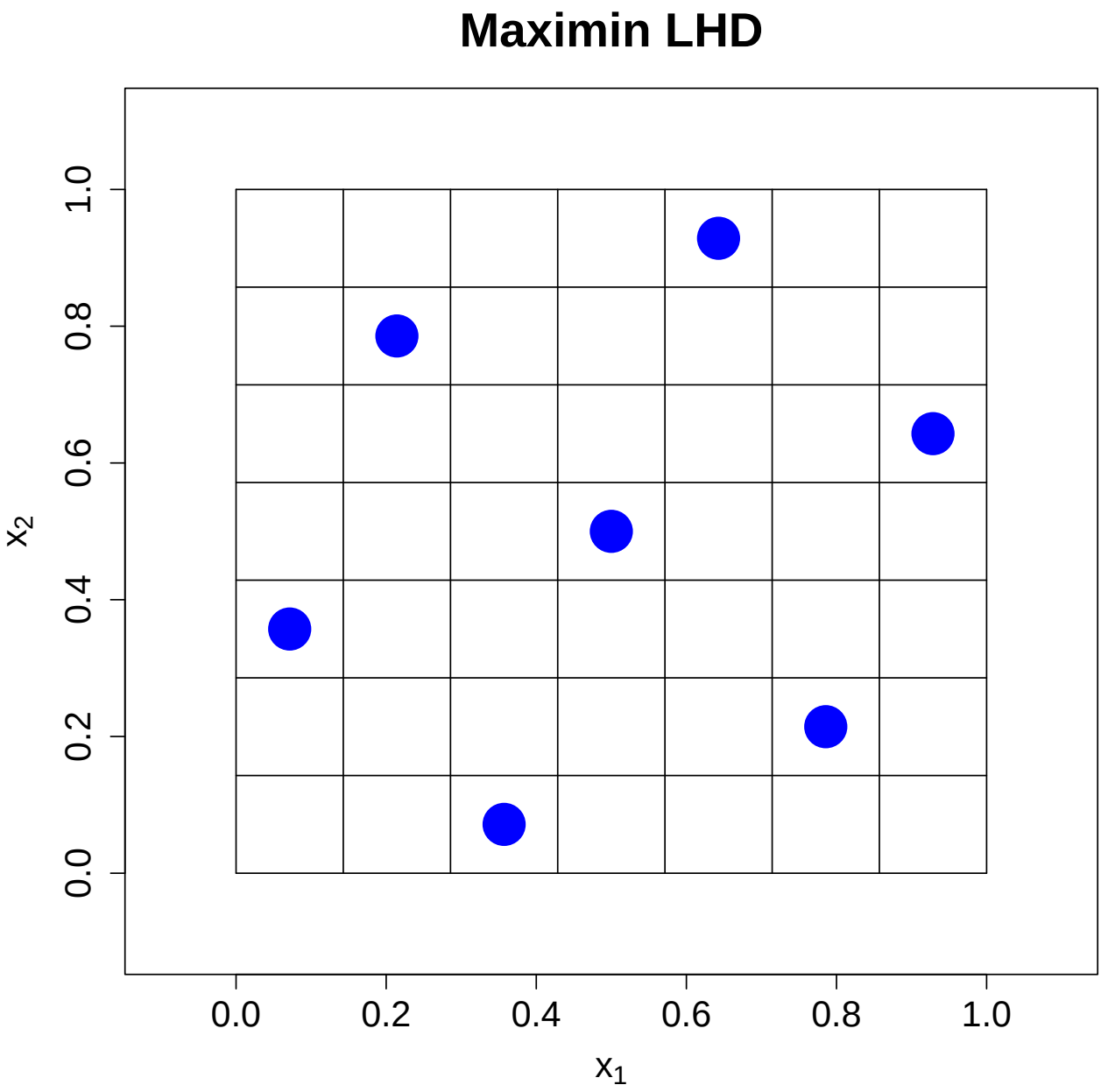
4 Latin Hypercube Designs

4.1 Maximin LHD

```
p <- 2
n <- 7
D <- maximinLHD(n, p)$design

plot(D, pch = 16, cex = 4, col = "blue", xlim = c(-0.1, 1.1), ylim = c(-0.1, 1.1),
     xlab = expression(x[1]), ylab = expression(x[2]),
     cex.lab = 1.5, cex.axis = 1.5, main = "Maximin LHD", cex.main = 2)
for (i in 1:(n+1)) {
  segments((i-1)/n, 0, (i-1)/n, 1)
```

```
  segments(0, (i-1)/n, 1, (i-1)/n)
}
```



4.2 MmLHD vs Chebyshev-MmLHD

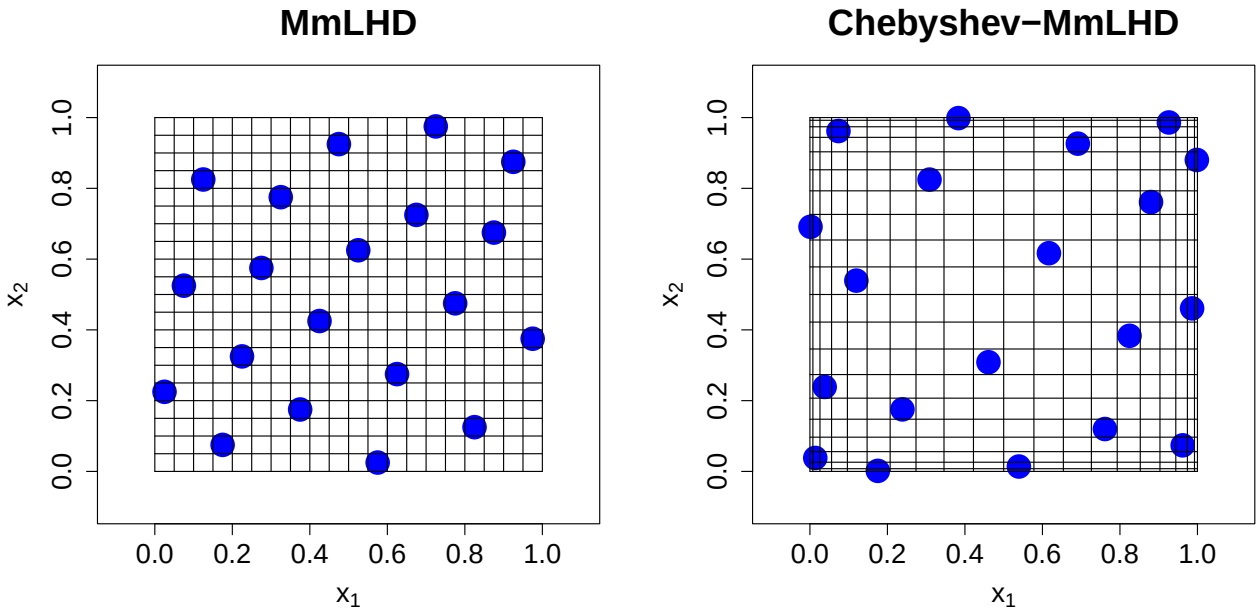
```
p <- 2
n <- 20
set.seed(1)
D <- maximinLHD(n, p)$design

par(mar = c(5, 5, 4, 2))
par(mfrow = c(1, 2))

plot(D, pch = 16, cex = 3, col = "blue", xlim = c(-0.1, 1.1), ylim = c(-0.1, 1.1),
```

```
     xlab = expression(x[1]), ylab = expression(x[2]),
     cex.lab = 1.5, cex.axis = 1.5, main = "MmLHD", cex.main = 2)
for (i in 1:(n+1)) {
  segments((i-1)/n, 0, (i-1)/n, 1)
  segments(0, (i-1)/n, 1, (i-1)/n)
}

D1 <- (1 + cos(pi * D)) / 2
plot(D1, pch = 16, cex = 3, col = "blue", xlim = c(-0.1, 1.1), ylim = c(-0.1, 1.1),
     xlab = expression(x[1]), ylab = expression(x[2]),
     cex.lab = 1.5, cex.axis = 1.5, main = "Chebyshev-MmLHD", cex.main = 2)
d1 <- sort(D1[, 1])
l1 <- c(0, (d1[-1] + d1[-n]) / 2, 1)
d2 <- sort(D1[, 2])
l2 <- c(0, (d2[-1] + d2[-n]) / 2, 1)
for (i in 2:(n+2)) {
  segments(l1[i-1], 0, l1[i-1], 1)
  segments(0, l2[i-1], 1, l2[i-1])
}
```




```
par(mfrow = c(1, 1))
```

5 MaxPro Designs

```
p <- 2

par(mar = c(5, 5, 4, 2))
par(mfrow = c(1, 2))

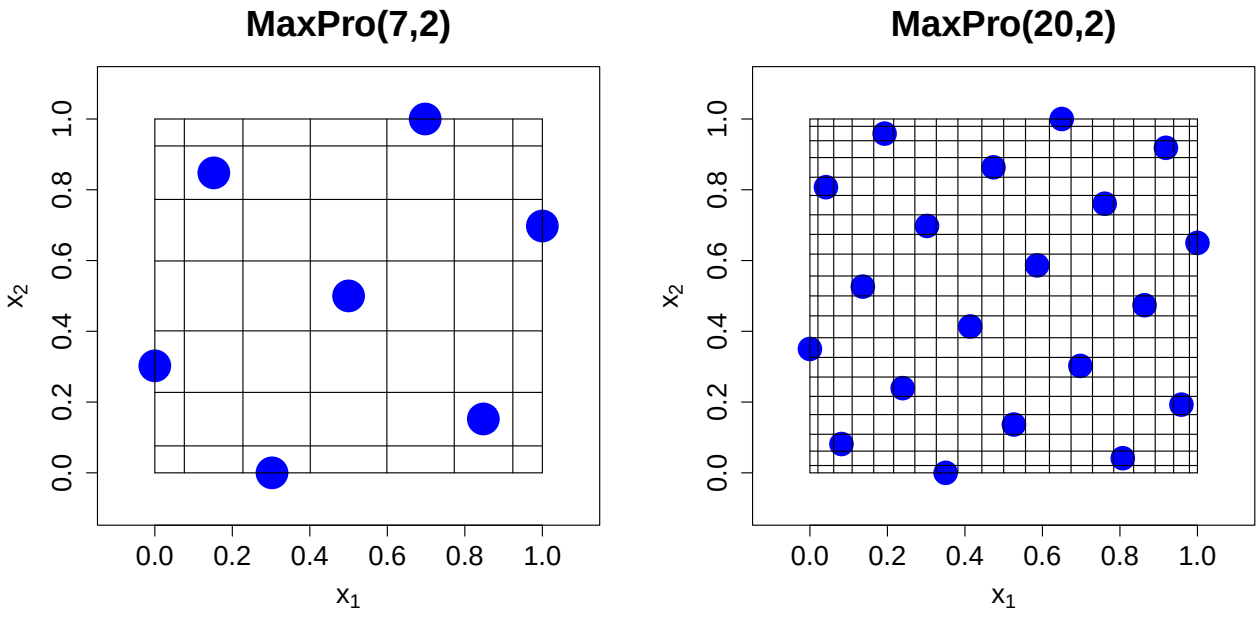
# MaxPro(7,2)
n <- 7
set.seed(10)
a <- MaxPro(MaxProLHD(n, p)$Design)
```

```r
D1 <- a$Design

plot(D1, pch = 16, cex = 4, col = "blue", xlim = c(-0.1, 1.1), ylim = c(-0.1, 1.1),
     xlab = expression(x[1]), ylab = expression(x[2]),
     cex.lab = 1.5, cex.axis = 1.5, main = "MaxPro(7,2)", cex.main = 2)
d1 <- sort(D1[, 1])
l1 <- c(0, (d1[-1] + d1[-n]) / 2, 1)
d2 <- sort(D1[, 2])
l2 <- c(0, (d2[-1] + d2[-n]) / 2, 1)
for (i in 2:(n+2)) {
  segments(l1[i-1], 0, l1[i-1], 1)
  segments(0, l2[i-1], 1, l2[i-1])
}

# MaxPro(20,2)
n <- 20
set.seed(10)
D <- maxpro.optim(maxproLHD(n, p)$design, iteration = 100)$design

plot(D, pch = 16, cex = 3, col = "blue", xlim = c(-0.1, 1.1), ylim = c(-0.1, 1.1),
     xlab = expression(x[1]), ylab = expression(x[2]),
     cex.lab = 1.5, cex.axis = 1.5, main = "MaxPro(20,2)", cex.main = 2)
d1 <- sort(D[, 1])
l1 <- c(0, (d1[-1] + d1[-n]) / 2, 1)
d2 <- sort(D[, 2])
l2 <- c(0, (d2[-1] + d2[-n]) / 2, 1)
for (i in 2:(n+2)) {
  segments(l1[i-1], 0, l1[i-1], 1)
  segments(0, l2[i-1], 1, l2[i-1])
}
```



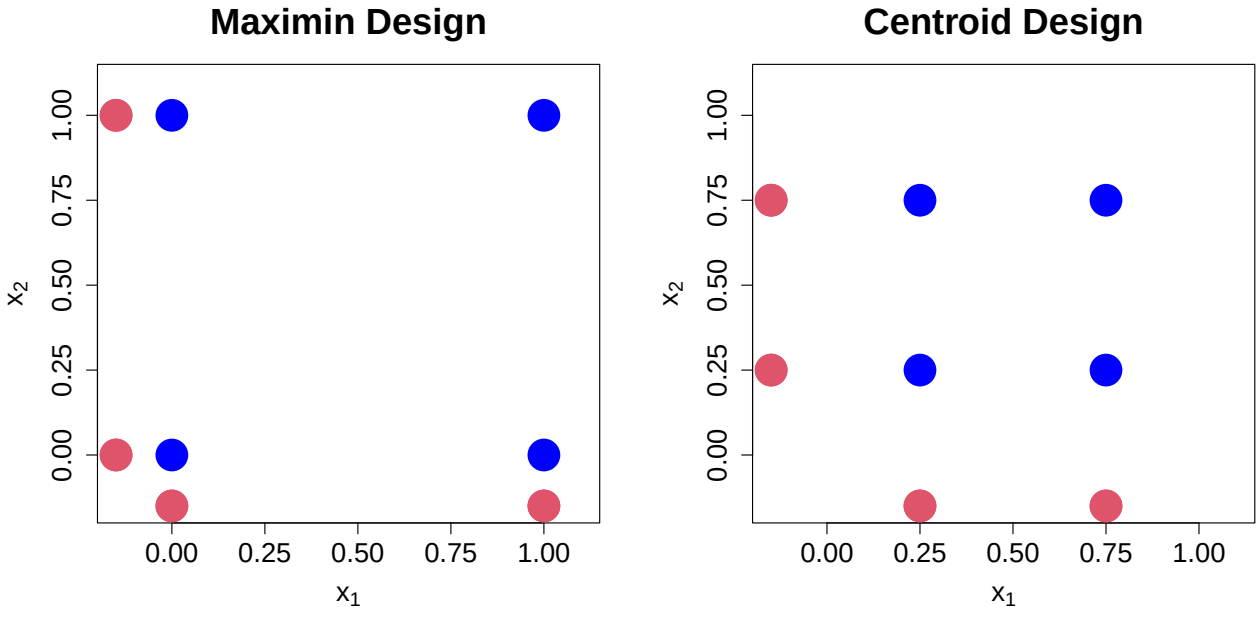

```r
par(mfrow = c(1, 1))
```

6 Projection Properties Comparison

```
par(mar = c(5, 5, 4, 2))
par(mfrow = c(1, 2))

# Maximin design
D <- matrix(c(0, 0, 0, 1, 1, 0, 1, 1), nrow = 4, ncol = 2, byrow = TRUE)
plot(D, pch = 16, xaxt = "n", yaxt = "n", cex = 4, col = "blue",
     xlim = c(-0.15, 1.1), ylim = c(-0.15, 1.1),
     xlab = expression(x[1]), ylab = expression(x[2]),
     cex.lab = 1.5, main = "Maximin Design", cex.main = 2)
axis(1, at = seq(0, 1, by = 0.25), cex.axis = 1.5)
axis(2, at = seq(0, 1, by = 0.25), cex.axis = 1.5)
points(cbind(D[, 1], -0.15), col = 2, pch = 16, cex = 4)
points(cbind(-0.15, D[, 2]), col = 2, pch = 16, cex = 4)

# Centroid design
D <- matrix(c(0.25, 0.25, 0.25, 0.75, 0.75, 0.25, 0.75, 0.75),
            nrow = 4, ncol = 2, byrow = TRUE)
plot(D, pch = 16, xaxt = "n", yaxt = "n", cex = 4, col = "blue",
     xlim = c(-0.15, 1.1), ylim = c(-0.15, 1.1),
     xlab = expression(x[1]), ylab = expression(x[2]),
     cex.lab = 1.5, main = "Centroid Design", cex.main = 2)
axis(1, at = seq(0, 1, by = 0.25), cex.axis = 1.5)
axis(2, at = seq(0, 1, by = 0.25), cex.axis = 1.5)
points(cbind(D[, 1], -0.15), col = 2, pch = 16, cex = 4)
points(cbind(-0.15, D[, 2]), col = 2, pch = 16, cex = 4)
```



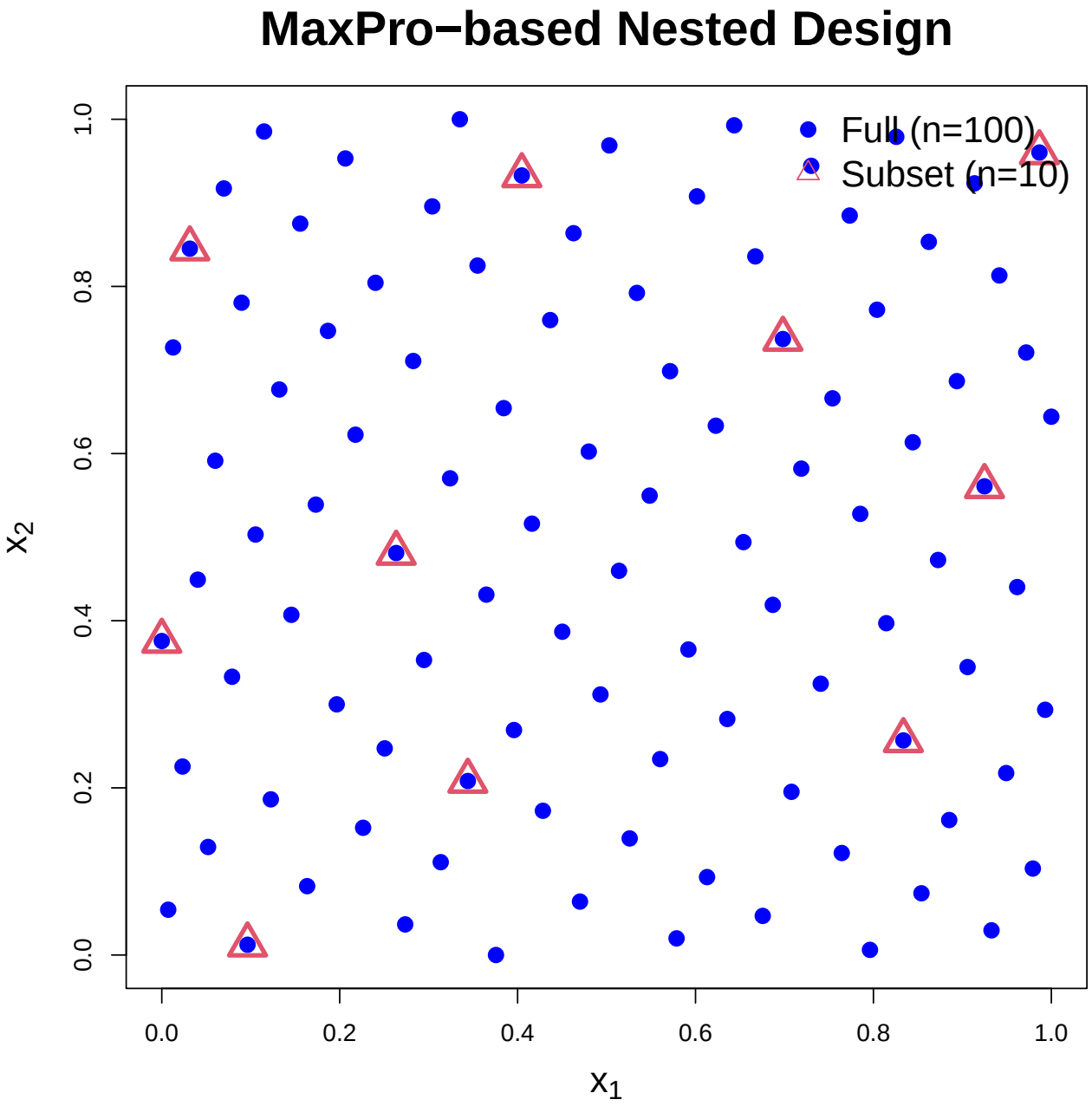

```
par(mfrow = c(1, 1))
```

7 Nested Designs

```
n1 <- 100
n2 <- 10
p <- 2

set.seed(1)
D1 <- maxpro.optim(maxproLHD(n1, p)$design)$design
D2 <- maxpro.remove(D1, n.remove = n1 - n2, delta = 1/n1^2)

par(mar = c(5, 5, 4, 2))
plot(D1, xlim = c(0, 1), ylim = c(0, 1), pch = 16, col = "blue", lwd = 2, cex = 1.5,
     main = "MaxPro-based Nested Design",
     xlab = expression(x[1]), ylab = expression(x[2]),
     cex.lab = 1.5, cex.main = 2)
points(D2, pch = 2, col = 2, lwd = 3, cex = 2.5)
legend("topright", legend = c("Full (n=100)", "Subset (n=10)"),
       pch = c(16, 2), col = c("blue", 2), cex = 1.5, bty = "n")
```



8 Summary

This chapter covered:

1. **Clustering-based designs** using Voronoi tessellation
2. **Maximin designs** maximizing the minimum distance between design points
3. **Latin hypercube designs (LHD)** with projection properties
4. **MaxPro designs** optimizing projection properties
5. **Nested designs** for multi-fidelity experiments

Note: Minimax designs are omitted as the ***`minimaxdesign`*** *package has been archived from CRAN.*